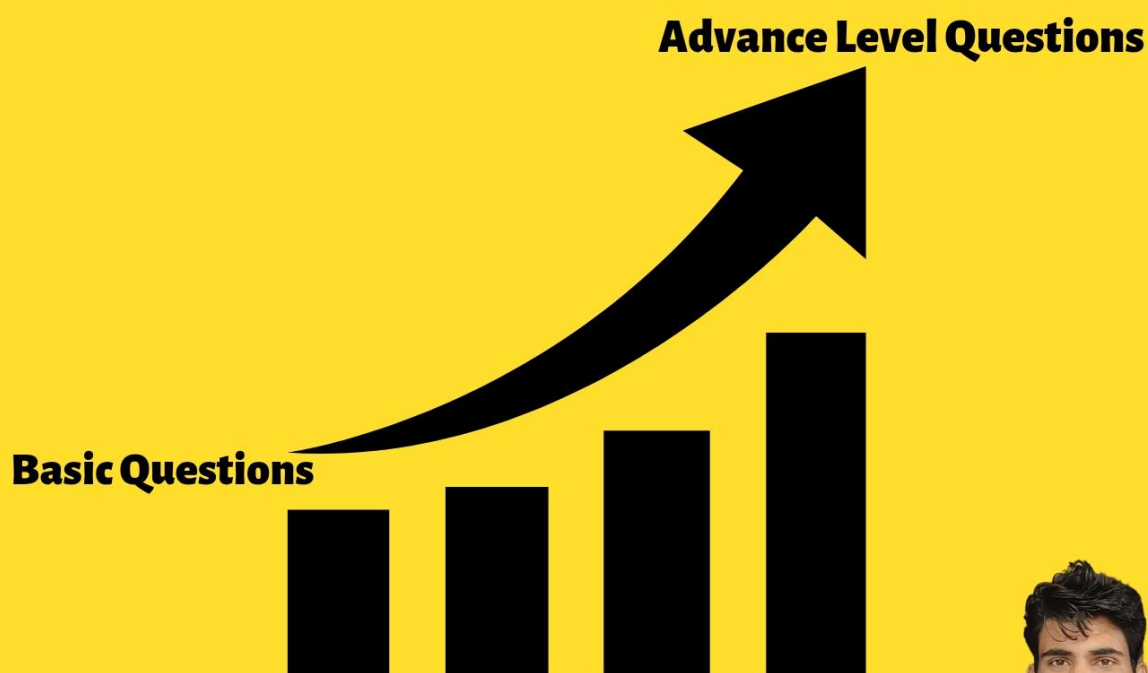


MATHS

LCM - NOTES

ALL TYPE WISE QUESTIONS



Maths with Gajanand



LCM

Least Common Multiple

12 $\rightarrow$ 24, 36, 48, 60, 72, 84, 96
15 $\rightarrow$ 30, 45, 60, 75, 90, 105, 120

LCM (12, 15) = 60 $\Leftarrow$ Least Common multiple of both

Methods of finding LCM

Find LCM of 24, 30, 40

m-I Basic

24 $\rightarrow$ 48, 72, 96, 120, 144
30 $\rightarrow$ 60, 90, 120, 150, 180
40 $\rightarrow$ 80, 120, 160, 200

LCM = 120 Ans

m-II

2	24, 30, 40
2	12, 15, 20
2	6, 15, 10
3	3, 15, 5
5	1, 5, 5
	1, 1, 1

LCM = $2 \times 2 \times 2 \times 3 \times 5 = 120$ Ans

Ex-III LCM (24, 30, 40)

$$24 \rightarrow 4 \times 6$$

$$30 \rightarrow 3 \times 10$$

$$40 \rightarrow 4 \times 10$$

$$\text{LCM} = 4 \times 10 \times 3$$

$$= \boxed{120}$$

Ex. LCM of 24, 30, 36

$$24 \rightarrow 12 \times \textcircled{2}$$

$$30 \rightarrow 6 \times \textcircled{5}$$

$$\textcircled{36} \rightarrow$$

$$36 \times 5 \times 2 = \boxed{360}$$

6, 12 is included in 36 so skip them

Ex. LCM of 51, 91, 119

$$51 \rightarrow 17 \times \textcircled{3}$$

$$91 \rightarrow \textcircled{13} \times 7$$

$$\textcircled{119} \rightarrow$$

$$\text{LCM} = 119 \times 3 \times 13$$

Ex. LCM of 15, 20, 25, 30

$$15 \rightarrow 15 \times 1$$

$$20 \rightarrow 10 \times \textcircled{2}$$

$$25 \rightarrow 5 \times \textcircled{5}$$

$$30$$

$$\text{LCM}$$

$$30 \times 2 \times 5$$

$$= \boxed{300}$$

$\Rightarrow$ [LCM of Prime Numbers $\Rightarrow$ their product]

Ex. $\text{Lcm}(7, 11, 13) = 7 \times 11 \times 13$

OR
[LCM of Co-Prime Numbers $\Rightarrow$ their product]

Ex. The product of two co-prime numbers is 117. Then their LCM is

- (a) 117 (b) 9 (c) 13 (d) 39

[1] The LCM of two prime numbers x and y ($x > y$) is 161. The value of $(3y - x)$:

Solⁿ:- $161 = 7 \times 23$
 $x = 23, y = 7$
 $3y - x = 3 \times 7 - 23$
 $= \boxed{-2}$ Ans

[2] A and B are two prime numbers such that $A > B$ and their LCM is 209. The value of $A^2 - B$ is
[SSC CGL-Pre-2022]

Solⁿ $209 = 19 \times 11$
 $\uparrow \quad \uparrow$
 $A \quad B$
 $A^2 - B = (19)^2 - 11$
 $= 361 - 11 = \boxed{350}$ Ans

[3] If the ratio of the two numbers is 2:3 and their LCM is 54, then the sum of the two number is:

Solⁿ:-
 $\begin{array}{cc} A & B \\ 2x & 3x \\ \times 3 \swarrow & \nwarrow \times 3 \\ 18 & 27 \\ \uparrow & \uparrow \\ + & \\ \boxed{45} & \text{Ans} \end{array}$
 $\text{LCM} = 2 \times 3 \times x = 54$
 $x = 9$

[4] The LCM of two multiples of 12 is 1056. If one of the number is 132, the other number is:

Soln:-

A	B	
$12x$	$12y$	$12xy = 1056$
$\downarrow$	$\downarrow \times 8$	$xy = 88$
132		$11y = 88$
$x = 11$	96	$y = \text{8}$
	<u>Ans</u>	

[5] If the least common multiple of two numbers, 1728 and K is 5184, then how many values of K are possible?

Soln:-

$$1728 \rightarrow 2^6 \times 3^3 \quad K \rightarrow 2^0, 2^1, 2^2, 2^3, 2^4, 2^5, 2^6$$

$$LCM = 5184 = 2^6 \times 3^4 \quad 3 \rightarrow 3^4$$

$$K = 7 \times 1 = \text{7}$$

[6] The LCM of four consecutive numbers is 60. The sum of the first two numbers is equal to the fourth number. What is the sum of four number?

Soln:-

$$x, x+1, x+2, x+3$$

$$x+x+1 = x+3$$

$$x = 2$$

$$2, +3, +4, +5 = \text{14} \quad \underline{\text{Ans}}$$

[7] LCM of three natural numbers 150, 144 and x is 10800. How many values of x are possible?

Soln:-

$$150 \rightarrow 2^1 \times 3^1 \times 5^2$$

$$144 \rightarrow 2^4 \times 3^2$$

$$LCM \rightarrow 10800 = 2^4 \times 3^3 \times 5^2$$

$$\begin{array}{rcl}
 x & = & 2^0 \times 3^3 \times 5^0 \\
 & & 2^1 \quad \quad \quad 5^1 \\
 & & 2^2 \quad \quad \quad 5^2 \\
 & & 2^3 \\
 & & 2^4 \\
 \hline
 & & 5 \times 1 \times 3 = \boxed{15} \text{ Ans}
 \end{array}$$

Type-1

* The least number which is exactly divisible by a, b, c
 $= \text{Lcm}(a, b, c)$

8. The least number which is exactly divisible by 18, 20, 24 is.

Soln:- $\text{Lcm}(18, 20, 24) = \boxed{360} \text{ Ans}$

9. The greatest number of 5 digit that is exactly divisible by each 8, 12, 15 and 20 is:

- (a) 99950 (b) 99940 (c) 99980 (d) 99960

Soln:- m-I Only option (d) is divisible by 8

m-II Basic

$$\text{Lcm}(8, 12, 15, 20) = 120$$

$$120 \overline{) 99999} (833$$

$$\underline{- 960}$$

$$399$$

$$\underline{- 360}$$

$$399$$

$$\underline{- 360}$$

Rem. $\rightarrow 39$

$$\text{No.} = 99999 - 39$$

$$= \boxed{99960} \text{ Ans}$$

10. The largest 4-digit number exactly divisible by each of 12, 15, 18 and 27 is:

- (a) 9690 (b) 9720 (c) 9930 (d) 9960

Soln:- Only (b) option is divisible by 9

$$\boxed{\text{m-II}} \quad \text{Lcm}(12, 15, 18, 27) = 27 \times 2 \times 5 \times 2 \\ = 540$$

$$\begin{array}{r} 540 \overline{) 9999} \quad (18 \\ \underline{540} \\ 4599 \\ \underline{4320} \\ \text{Rem} \rightarrow 279 \end{array}$$

$$\begin{aligned} \text{No.} &= 9999 - 279 \\ &= \boxed{9720} \quad \underline{\text{Ans}} \end{aligned}$$

11. The least number of 5-digit which exactly divided by 18, 20 and 24 ?

Soln:-

$$\begin{array}{r} 360 \overline{) 10000} \quad (27 \\ \underline{- 720} \\ 2800 \\ \underline{- 2520} \\ \text{Rem.} \rightarrow 280 \end{array}$$

$$360 - 280 = 80$$

$$\begin{aligned} \text{No.} &= 10000 + 80 \\ &= \boxed{10080} \quad \underline{\text{Ans}} \end{aligned}$$

12. The least number of 5-digit which exactly divided by 52, 56, 78 and 91 ?

- (a) 10290 (b) 10860 (c) 10920 (d) 10580

Soln:- Only option (c) divisible by 8

[13] The number between 4000 and 5000 that is divisible by each of 12, 18, 21 and 32 is

- (a) 4203 (b) 4023 (c) 4032 (d) 4302

Soln:- m-I by option method:-

4032 is Only divided by 4

m-II Basic:-

$$\text{Lcm}(12, 18, 21, 32) = 2016$$

$$2016 \times k = \boxed{4032}$$

$\downarrow$
2

[14] What least value which should be added to 1812 to make it divisible by 7, 11, and 14?

- (a) 12 (b) 36 (c) 72 (d) 154

Soln:- by option method = $1812 + 36 = 1848$
divisible by 7, 11, 14

m-II Basic:-

$$\text{Lcm}(7, 11, 14) = 154$$

$$\begin{array}{r} 154 \overline{) 1812} \quad (11 \\ - 154 \\ \hline 272 \\ - 154 \\ \hline 118 \end{array}$$

Rem $\rightarrow 118$

$$\text{No.} = 154 - 118 = \boxed{36} \quad \underline{\text{Ans}}$$

[15] What least value which should be subtract to 10823 to make it divisible by 18, 20, 24?

Soln $\text{Lcm}(18, 20, 24) = 360$

$$\begin{array}{r} 360 \overline{) 10823} \\ - 10800 \\ \hline 23 \end{array}$$

Rem. $\rightarrow 23$

$$\text{No.} = \boxed{23} \quad \underline{\text{Ans}}$$

- 16] The least number, which is a perfect square and is divisible by each of the number 16, 20 and 24 is:
- (a) 1600 (b) 3600 (c) 6400 (d) 14400

Soln Only ^(least) 3600 is divisible by 8, 3
(14400 is also divisible by 8, 3 but it is greater than 3600)

m-II] $\text{Lcm}(16, 20, 24) = 2^4 \times 5 \times 3$
 $= 2^4 \times 3 \times 5 \leftarrow \text{make even power for perfect square}$

So multiply 3×5

$$= 2^4 \times 3^2 \times 5^2$$
$$= \boxed{3600} \text{ Ans}$$

Type-2

* [The no. when divided by a, b, c leaves remainder 'r' in each case $\Rightarrow \boxed{\text{Lcm}(a, b, c) \times k + r}$]

- 17] Find the least number which divided by 18, 20 and 24 leaves 3 as remainder in each case:

Soln:-

$$\boxed{\text{Lcm}(a, b, c) + r}$$

$$\text{Lcm}(18, 20, 24) = 360$$

$$\text{No.} = 360 + 3$$

$$= \boxed{363} \text{ Ans}$$

[18] The greatest number of 4 digits which gives remainder 3 when divided by 18, 20, 24 is:

Soln Basic

$$\begin{array}{r} 360 \overline{) 9999} \begin{array}{l} 27 \\ 720 \\ \hline 2799 \\ - 2520 \\ \hline \end{array} \end{array}$$

Rem $\rightarrow 279$

$$9999 - 279 = 9720$$

$$\text{No.} = 9720 + 3$$

$$= \boxed{9723} \text{ Ans}$$

[19] The greatest number of six digits which gives remainder 10 when divided by 36, 48, 54, 60 and 90 is:

Soln (a) 997930 (b) 997240 (c) 997910 (d) 999908

After subtracting 10 from all options, only (a) is divisible by 9

[20] The smallest number, which when increased by 5 is divisible by each of 24, 32, 36 and 54, is:

Soln (a) 869

$$\begin{array}{r} +5 \\ 874 \\ \hline \end{array} \neq 9$$

(b) 859

$$\begin{array}{r} +5 \\ 864 \\ \hline \end{array} \div 9 \checkmark$$

(c) 541

$$\begin{array}{r} +5 \\ 546 \\ \hline \end{array} \neq 9$$

(d) 827

$$\begin{array}{r} +5 \\ 832 \\ \hline \end{array} \neq 9$$

[11] $\text{Lcm}(a, b, c) - 5$

$$= \text{Lcm}(24, 32, 36, 54) - 5$$

$$= 864 - 5$$

$$= \boxed{859} \text{ Ans}$$

- Q1** Let x be the smallest number, which when added to 2000 makes the resulting number divisible by 12, 16, 18 and 21. The sum of the digits of x is
- (a) 6 (b) 5 (c) 7 (d) 5

Soln:- **m-I** $\text{LCM}(12, 16, 18, 21) = 21 \times 6 \times 8$
 $= 1008 \xrightarrow{\times 2} 2016$
 $\downarrow$
 $2000 + \boxed{16} \xrightarrow{+} 7 \text{ Ans}$

m-II
 $2000 + x \Rightarrow 12, 16, 18, 21$
 $\downarrow$
 9×2
 $(2000 + x) \text{ will be div. by } 9$

$2000 + x \rightarrow \text{digit sum} = 9$
 $\downarrow$
 $2 + x = 9$
 $x = \boxed{7} \text{ Ans}$

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- Q2** What least number must be subtracted from 1936 so that the resulting number when divided by 9, 10 and 15 will leaves in each case the same remainder?

Soln:- (a) 37 (b) 36 (c) 39 (d) 30 (7)

by option (c) satisfy the condition

Basic $\text{LCM}(9, 10, 15) = 90$

$90 \overline{) 1936}$
 $\underline{ 1800}$
 $ 136$
 $\text{Rem} = 46$

$\text{No.} = 46 - 7$
 $= \boxed{39} \text{ Ans}$

Q3 A number between 1000 and 2000 which when divided by 30, 36 and 80 gives a remainder 11 in each case is:

(a) 11523
m-I $\begin{array}{r} -11 \\ 11512 \end{array}$ $\div 9$

(b) 1451
 $\begin{array}{r} -11 \\ 1440 \end{array}$ $\div 9$ ✓

(c) 1641
 $\begin{array}{r} -11 \\ 1630 \end{array}$ $\div 9$ ✗

(d) 1712
 $\begin{array}{r} -11 \\ 1701 \end{array}$ $\div 9$ ✓
 $\div 2$ ✗

m-II $\text{LCM}(30, 36, 80) = 720$

No. = $720K + 11$

$K=2 \Rightarrow 1440 + 11 = \boxed{1451}$ Ans

Q4 Let x be the least number, which divided by 5, 6, 7 and 8 leaves a remainder 3 in each case but when divided by 9 leaves remainder 0, the sum of digits of x is: (a) 17 (b) 18 (c) 19 (d) 20

Soln:- $\text{LCM}(5, 6, 7, 8) = 840$

No. = $\frac{840n + 3}{9} \Rightarrow \frac{3n + 3}{9}$

Take $n=2$

= $840 \times 2 + 3 = \boxed{1683}$ Ans

digit sum = $\boxed{18}$

m-II Only (b) has digit sum $\boxed{9}$

Q5 The least multiple of 13, which on dividing by 4, 5, 6, 7 and 8 leaves remainder 2 in each case is

(a) 2520

(b) 842

✓ (c) 2522

(d) 840

Soln:- Only (c) option is divisible by 13

m-II $\text{LCM}(4, 5, 6, 7, 8) = 840$

required No. = $840 \times k + 2$

$\frac{840k + 2}{13} = \text{rem } 0$

$\boxed{k=3}$

= $840 \times 3 + 2$

= $2520 + 2 = \boxed{2522}$ Ans

Q6 The least multiple of 7, which leaves the remainder 4, when divided by any of 6, 9, 15 and 18, is ?
 (a) 76 (b) 94 (c) 184 ✓(d) 364

Soln:- Only (d) option is divisible by 7

m-II $\text{Lcm}(6, 9, 15, 18) = 90$

Required No. = $90 \times k + 4$

$$\frac{90k+4}{7} = \frac{6k+4}{7} \quad = 90 \times 4 + 4$$

$$= 360 + 4 = \boxed{364} \text{ Ans}$$

$k=4$

Q7 A number x is divisible by 7 when this number is divided by 8, 12 and 16. It leaves a remainder 3 in each case. The least value of x is:

Soln:- (a) 148 (b) 149 (c) 150 ✓(d) 147

m-I Only (d) option is divisible by 7

m-II $\text{Lcm}(8, 12, 16) = 16 \times 3$
 $= 48$

Required No. = $48 \times k + 3$

$$\frac{48k+3}{7} = \frac{6k+3}{7} \quad = 48 \times 3 + 3$$

$$k=3 \quad = \boxed{147} \text{ Ans}$$

Type-3

The Number which when divided by a, b, c respectively gives remainder r_1, r_2 and r_3 such that

*

$$\text{Common difference (d)} = a - r_1 = b - r_2 = c - r_3$$

$$\text{Number} = \boxed{\text{Lcm}(a, b, c)k - d}$$

[28] The least number, which when divided by 18, 27 and 36 separately leaves remainder 5, 14, 23 respectively is

- (a) 95 (b) 113 (c) 49 (d) 77

Soln:-

$$d = 18 - 5 = 27 - 14 = 36 - 23 = \boxed{13}$$

$$\text{Lcm}(18, 27, 36) = 108$$

$$\text{Required No} = 108 - 13 = \boxed{95}$$

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[29] Find the least multiple of 23, which when divided by 18, 21 and 24 leaves the remainder 7, 10 and 13 respectively

- (a) 3013 (b) 3024 (c) 3002 (d) 3036

Soln:-

$$d = 18 - 7 = 11$$

$$21 - 10 = 11$$

$$24 - 13 = 11$$

$$\begin{aligned} \text{Lcm}(18, 21, 24) &= 24 \times 7 \times 3 \\ &= 504 \end{aligned}$$

$$\text{Required No.} = 504k - 11$$

$$\frac{504k - 11}{23} = 21k - 11$$

$$\boxed{k=6}$$

$$= 504 \times 6 - 11$$

$$= \boxed{3013} \text{ Ans}$$

[30] The least number of 6-digit which when divide by 15, 25, 35, 42 and 70 respectively, leaves the remainder 11, 21, 31, 38 and 66 respectively ?

- (a) 100796 (b) 100256 (c) 100446 (d) 101046

Soln:-

$$D = 15-11 = 25-21 = 35-31 = 4$$

$$\text{Lcm}(15, 25, 35, 42, 70) = 1050$$

$$\text{Required No.} = 100800 - 4$$

$$= \boxed{100796}$$

$$\begin{array}{r} 1050 \overline{) 100000} \quad 95 \\ \underline{9450} \\ 5500 \\ \underline{- 5250} \\ 250 \end{array}$$

200

$= 100000 + 800 = 100800$

[m-II] Option method :- After adding 4 only option (a) is divisible by 7

[31] The greatest number of four digits which when divided by 3, 5, 7, 9 leave remainder 1, 3, 5, 7 respectively.

- (a) 9763 (b) 9764 (c) 9766 (d) 9765

Soln:-

$$d = 3-1 = 5-3 = 7-5 = 9-7 = 2$$

$$\text{Lcm}(3, 5, 7, 9) = 9 \times 7 \times 5 = 315$$

$$\begin{array}{r} 315 \overline{) 9999} \quad 31 \\ \underline{945} \\ 549 \\ \underline{315} \\ 234 \end{array}$$

$$\text{Rem} = 234$$

$$\text{Required No.} = 9765 - 2$$

$$= \boxed{9763}$$

- [32] Five bells ring together at the intervals of 3, 5, 8, 9 and 10 seconds. All the bells ring simultaneously at the same time. They will again ring simultaneously after:
- (a) 6 min (b) 8 min (c) 9 min (d) 4 min.

Soln:- $\text{Lcm}(3, 5, 8, 9, 10) = 360 \text{ Seconds}$
 $= \boxed{6 \text{ min.}}$ after

- [33] Three electronic devices make a beep after 48 seconds, 72 seconds and 108 seconds respectively, they all beeped together at 10 am in the morning. next time at which time they will together beep?

- (a) 10:07:12 (b) 10:07:24 (c) 10:07:36 (d) 10:07:48

[SSC-CGL - 2017]

Soln:- $\text{Lcm}(48, 72, 108) = 432 \text{ second}$
 $\Downarrow$
 7 min 12 sec.

$\Rightarrow 10 \text{ am} + 7 \text{ min. } 12 \text{ sec}$

$\Rightarrow \boxed{10:07:12}$ Ans

- [34] A, B, C begin to run in the same direction from the same point in a circular stadium at the same time. A completes one round in 36 seconds, B in 48 seconds and C in 60 seconds. In how much time will they meet again at the initial point?

Soln:- $\text{Lcm}(36, 48, 60) = 60 \times 4 \times 3$
 $= 720$
 $= \boxed{12 \text{ min}}$ Ans

35 Three man step off together from same spot. Their steps measure 63cm, 70cm and 77cm respectively. The minimum distance each should cover so that all can cover the distance in complete steps is:

Soln:- $\text{LCM}(63, 70, 77) = 7 \times 11 \times 10 \times 9 = \boxed{6930}$ Ans

36 What is the least number of soldiers that can be drawn up in troops of 10, 12, 15, 18 and 20 soldiers and also in form of a solid square?

[SSC CPO - 2020]

Soln:- $\text{LCM}(10, 12, 15, 18, 20) = 180$

Perfect square = $180 \times 5 = \boxed{900}$ Ans

